Feb 24 2026

Lasalle-Krasovskii Invariance Principle

- Time invariant systems
- Allow to conclude AS from $\dot{V}(x) \leq 0$ if additional conditions hold

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Suppose

$$\Omega_c = \{x : V(x) \leq c\} \text{ is bounded and } \dot{V}(x) \leq 0 \text{ in } \Omega_c$$

Using Lyapunov stability theory, we could already conclude that:

if $x(0) \in \Omega_c$

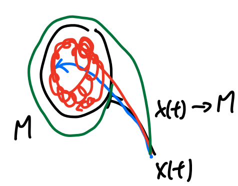
then $x(t) \in \Omega_c$ $\quad V(t) \to \tilde{c} \leq c$

and therefore $x(t) \to \Omega_{\tilde{c}}$

$S := \{x \in \Omega_c : \dot{V}(x) = 0\}$

M is the largest invariant set in S

$x(t) \to M \quad \forall x(0) \in \Omega_c$



Ex. $\quad \dot{y} = ay + u$

$u = -ky \quad , \dot{k} = \gamma y^2 \quad , \gamma > 0$

$\dot{y} = ay - ky$

$\dot{k} = \gamma y^2$

$x_1 = y, x_2 = k$

$\dot{x}_1 = f_1(x_1, x_2) = a x_1 - x_1 x_2$

$\dot{x}_2 = f_2(x_1, x_2) = \gamma x_1^2$

Consider the following Lyapunov function candidate

$V(x) = \frac{1}{2} x_1^2 + \frac{1}{2\gamma}(x_2 - b)^2 \qquad b > a$

$$\dot{V}(x) = x_1 \dot{x}_1 + \frac{1}{\gamma}(x_2 - b)\dot{x}_2$$

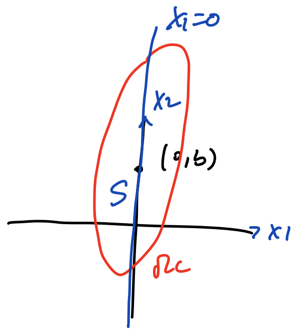
$$= x_1(ax_1 - x_1 x_2) + \frac{1}{\gamma}(x_2 - b)(\gamma x_1^2)$$

$$= a x_1^2 - b x_1^2$$

$$= (a-b)x_1^2 \le 0 \qquad (b > a)$$

$(0, b)$ is stable

<u>LaSalle</u>



$\dot{V} = 0$

$M = S$

$x(t) \to M$ (x_2-axis)

$x_1(t) \to 0$

Back to linear system $\dot{x} = Ax$

<u>Recall</u>

$x = 0$ is stable if $Re\{\lambda_i(A)\} \le 0 \quad \forall i$

and the e-values on the imaginary axis have Jordan blocks of order one.

If $Re\{\lambda_i(A)\} < 0 \Rightarrow$ AS

(Hurwitz)

$V(x) = x^T P x$, $P = P^T > 0$

$$\dot{V}(x) = x^T P \dot{x} + \dot{x}^T P x$$

$$= x^T P A x + x^T A^T P x$$

$$= x^T (PA + A^T P) x \quad = -x^T Q x$$

$$(PA + A^T P)^T = A^T P^T + P^T A$$
$$= A^T P + PA$$

$$A^T P + PA = -Q$$

$$P = \int_0^\infty e^{A^T t} Q e^{At}\, dt$$

- $P > 0$

$$x^T P x = \int_0^\infty \underbrace{x^T e^{A^T t}}_{y^T} Q \underbrace{e^{At} x}_{y := e^{At} x}\, dt$$

- $A^T P + PA = -Q$

$$= \int_0^\infty \underbrace{\left(A^T e^{A^T t} Q e^{At} + e^{A^T t} Q e^{At} A\right)}_{= \frac{d}{dt}\left(e^{A^T t} Q e^{At}\right)} dt$$

$$\frac{d}{dt}\left(e^{At}\right) = \frac{d}{dt}\left(I + At + \frac{A^2 t^2}{2!} + \cdots\right)$$
$$= A + A^2 t + \frac{A^3 t^2}{2!} + \cdots$$
$$= A\left(I + At + \frac{A^2 t^2}{2!} + \cdots\right)$$
$$= A e^{At}$$
$$= e^{At} A$$

$$\frac{d}{dt}\left(e^{A^T t} Q e^{At}\right) = e^{A^T t} Q \frac{d}{dt}\left(e^{At}\right) + \frac{d}{dt}\left(e^{A^T t}\right) Q e^{At}$$
$$= e^{A^T t} Q e^{At} A + A^T e^{A^T t} Q e^{At} \qquad \boxed{\text{QED}}$$

Therefore $A^T P + PA = e^{A^T t} Q e^{At} \Big|_0^\infty$

$$= 0 - Q$$
$$= -Q$$